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Effect of Interstitial Elements on Hall–Petch Coefficient of Ferritic Iron

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Yield strength was investigated in iron with different grain size and different contents of carbon and nitrogen. The Hall–Petch coefficient; k_y is originally very small in high purity iron (about 100 MPa·μm$^{1/2}$) but it increases with increasing the amount of solute carbon content. The value of k_y is enlarged to around 550 MPa·μm$^{1/2}$ by 60 ppm of solute carbon and levels off at around 600 MPa·μm$^{1/2}$ in the region above 60 ppm solute carbon. On the other hand, nitrogen hardly influences the k_y value. The mechanism of the change in Hall–Petch coefficient was then discussed in terms of the grain boundary segregation of carbon and nitrogen atoms.

KEY WORDS: ferritic iron; Hall–Petch coefficient; carbon; nitrogen; yield strength.

1. Introduction

In polycrystalline metals, yield strength increases with grain refinement in proportion to the inverse of square root of grain size, so called Hall–Petch relationship. In the case of polycrystalline iron, the following relationship has already been established for yield strength (σ_y) in the grain size region down to 0.2 μm,[1,2)] where "d" denotes the grain size in micrometer.

$$\sigma_y\,[\mathrm{MPa}] = 100 + 600 \times d\,[\mu\mathrm{m}]^{-1/2} \quad \text{.....(1)}$$

After that, many challenges for ultra grain refinement to 1 μm have been performed in conventional low carbon steel and it was confirmed that most of data are on the line[3)] drawn by the above equation. As far as discussing the grain size dependency of conventional low carbon steel, it is concluded that the Hall–Petch coefficient (k_y) should be the value of around 600 MPa·μm$^{1/2}$. On the other hand, it has recently reported that the k_y is very small (about 100 MPa·μm$^{1/2}$) in IF-steel.[4,5)] This suggests a certain impurity dependency of k_y in polycrystalline iron.

In this study, the carbon and nitrogen contents dependence of Hall–Petch coefficient of ferritic iron was investigated by using various irons containing different contents of carbon and nitrogen up to 60 ppm C or N. The mechanism of the change in Hall–Petch coefficient was then discussed in terms of the grain boundary segregation of carbon and nitrogen atoms.

2. Experimental Procedure

The chemical compositions of materials used in this investigation are listed in **Table 1**. These materials are conventional pure iron containing different amount of carbon and nitrogen up to 60 ppm in total. The symbol of each sample denotes the kind of element and its amount (ppm) added intentionally, for example, C60 means the iron containing about 60 ppm of carbon in total. In IF-steel, sufficient amount of Ti is added to fix carbon and nitrogen as Ti(C, N).

Ingots (25 kg) of 110 mm square in cross section were homogenized at 1 523 K for 3.6 ks and then hot-rolled to 10 mm thick. The hot-rolled plates were subjected to 90%

Table 1. Chemical compositions of polycrystalline ferritic steels used in this study (mass ppm).

	C	N	Si	Mn	P	S	Al	O	Ti	Fe
Base	5	9	<30	<30	<20	<3	40	21	<20	bal.
C30	28	11	<30	<30	<20	<3	30	51	—	bal.
C60	56	11	<30	<30	<20	<3	40	39	—	bal.
N30	11	24	<30	370	<20	<3	50	24	—	bal.
N60	5	54	<30	830	<20	<3	40	21	—	bal.
C30N30	35	26	<30	380	<20	<3	50	21	—	bal.
IF-steel	4	10	<30	<30	<20	<3	40	23	240	bal.

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cold rolling. Plates (150 mm×150 mm) cut from cold rolled materials were annealed at 973 K for 60 s through 600 s to change the ferrite grain size and then quenched into water to suppress the precipitation of carbide and nitride on cooling. The specimens were put in a thermostatic chamber set at 225 K soon after the annealing to suppress strain hardening. The grain size (nominal grain size) was measured by quadrature method.[6] Tensile testing was carried out at initial strain rate of $10^{-3}\,s^{-1}$ for plate test pieces standardized by JIS13B. In this article, the yield strength of specimens was defined as the upper-yielding stress or the elastic limit when the specimen exhibits continuous yielding. The elastic limit was estimated at the critical stress where the stress–strain curve begins to deviate from the linear relation of Young's modulus measured by resonance method. The yield strengths obtained were plotted against the inverse of square root of grain size on the basis of Hall–Petch relationship, and then the k_y was estimated by least-square method.

3. Results and Discussion

3.1. Microstructural Character of Specimens

Figure 1 displays the estimated amount of solute carbon and nitrogen. The amount of carbide and nitride was obtained by means of electrolytic extraction method, thus the amount of solute carbon and nitrogen was estimated by subtracting it from the total amount. However, from the difficulty of extraction of very tiny particles of carbide and nitride, the amount of solute carbon and nitrogen seems to be overestimated. The true amounts of solute carbon and nitrogen in IF-steel should be much smaller. In the following chapter, the solute carbon/nitrogen contents were used for data analysis when the effects of carbon/nitrogen on grain refinement strengthening were discussed.

The microstructure analysis by electron back scattering pattern is one of the useful methods for grain size evaluation because the informations on orientation distribution and grain size distribution can be obtained at the same time. For example, **Fig. 2** displays the orientation imaging map and the orientation distribution of crystal grains obtained in the sample C60 annealed at 973 K for 90 s and IF-steel annealed at 973 K for 120 s. The grain boundary in the imaging maps represents the high angle boundary with misorientation angle more than 15°. These results indicate random distribution of crystal grains. From the imaging maps, the nominal grain size was evaluated at 27 μm and 30 μm in the sample (a) and (c), respectively. The same microstructure analysis was performed for the other samples and the nominal grain size was evaluated objectively in each sample.

3.2. Effects of Carbon and Nitrogen on the Hall–Petch Coefficient

Figure 3 shows the relation between the yield strength measured by tensile testing and the inverse of square root of grain size. In all C-series (a) and N-series (b), the yield strength linearly increases with increasing the inverse of square root of grain size in accordance with Hall–Petch relationship. The equation expressing Hall–Petch relationship for each specimen is shown in the figure. It should be noted here that the slope of linear relation, Hall–Petch coefficient (k_y), is varied depending on the kind of interstitial elements

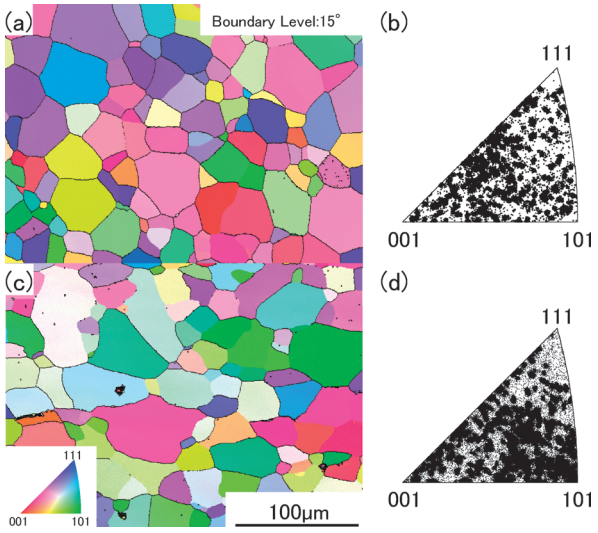


Fig. 2. Orientation imaging maps and inverse pole figures of C60 sample annealed at 973 K for 90 s ((a) and (b)) and of IF-steel annealed at 973 K for 120 s ((c) and (d)).

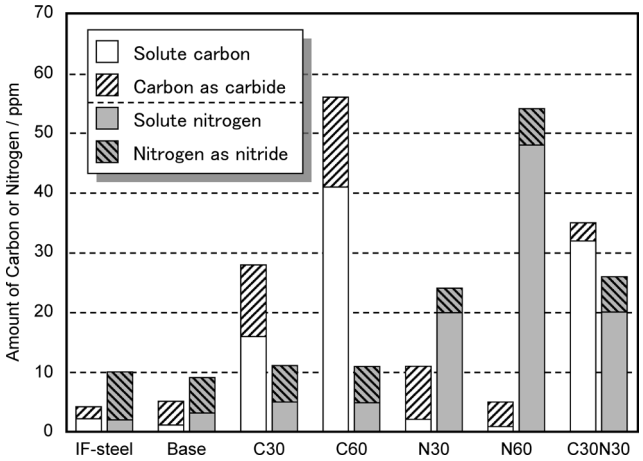


Fig. 1. Solute carbon and nitrogen estimated from the results of electrolytic extraction.

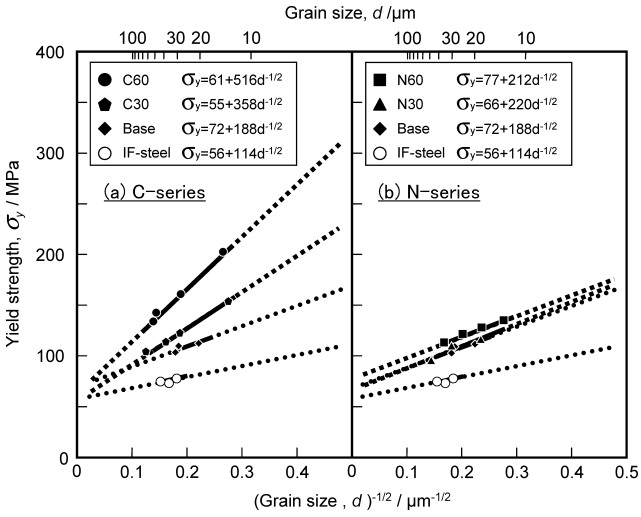


Fig. 3. Hall–Petch relationship in C-series (a) and N-series samples (b).

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and its content. The k_y of C-series monotonously increases with increasing solute carbon content (a), while that of N-series hardly changes even when the solute nitrogen content is increased more than 9 ppm (the content in BASE) (b). The IF-steel has the lowest k_y of all the specimens. From these results, it is suggested that the Hall–Petch coefficient strongly depends on the solute carbon content but not on the solute nitrogen content. On the other hand, the frictional stresses, σ_0, given by the extrapolation of the Hall–Petch relationship to infinitely large grain size, almost converge on one similar value around 60 MPa irrespective of carbon and nitrogen content. This corresponds to the yield strength of pure iron with infinitely large grain size given by the Pickering's prediction relation on yield strength of steel.[7] Since the σ_0 is the inherent strength without grain refinement strengthening, it is deduced that the solid solution strengthening by carbon and nitrogen would have a negligible small influence on the yield strength of ferritic iron. The change in k_y is shown in **Fig. 4** as a function of solute carbon or nitrogen content. The figure (a) indicates that the value of k_y increases significantly in the small carbon content region and reaches around 550 MPa·μm$^{1/2}$ by 60 ppm of solute carbon. As for the conventional low carbon steels, the value of k_y is 550–600 MPa·μm$^{1/2}$.[3] This result indicates that a small difference in solute carbon content causes the scattering of yield strength of low carbon steel through a marked change in k_y. On the other hand, the figure (b) demonstrates that solute nitrogen hardly influences to the k_y. When the influence of nitrogen is discussed on the grain size dependence, the effect of carbon should be eliminated in the N-series samples because these samples contain small amount of solute carbon as shown in Fig. 1. The difference from the value of IF-steel is probably due to the contribution of solute carbon.

Figure 5 summarizes all of data in the relation between k_y and solute carbon content. Data of the specimen used for the previous research[1,2] is added in this figure. This figure displays a clear tendency of leveling off of k_y at 550–600 MPa·μm$^{1/2}$. This means that solute carbon affects k_y greatly in the region below 60 ppm but does not give a large influence in the region above 60 ppm. It is confirmed again that k_y depends only on the amount of solute carbon regardless of solute nitrogen content. The change of k_y is approximately represented by the following equation as a function of solute carbon content.

$$k_y\,[\mathrm{MPa}\cdot\mu\mathrm{m}^{1/2}] = 100 + 120 \times [\mathrm{C}_{\mathrm{ppm}}]^{0.31} \quad \text{.....(2)}$$

3.3. Change in Hall–Petch Coefficient by Grain Boundary Segregation of Interstitial Alloy Element

The change in k_y due to interstitial alloy element has been well known in nitrogen bearing austenitic steels. Norström[8] firstly reported that the k_y of Fe–18%Cr–14%Ni–3%Mo alloy was increased by the addition of nitrogen. Authors[9] also systematically investigated the k_y of austenitic steels with different amounts of nitrogen, and revealed the relation between k_y and nitrogen concentration up to 1.2 mass%. The contribution of nitrogen to k_y has often been discussed since the Norström's report, and several mechanisms have been proposed. They could be classified to the following four ideas:

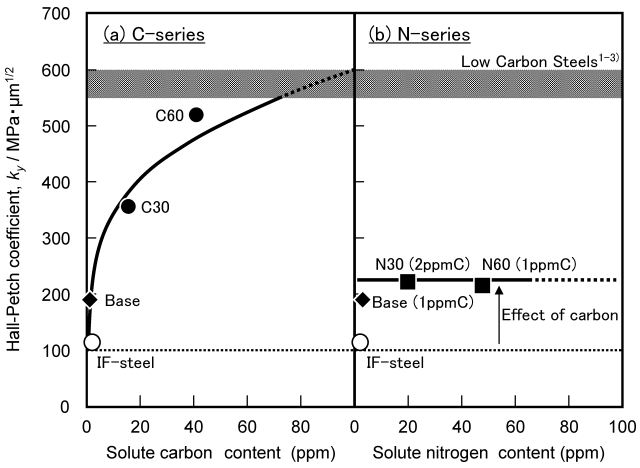


Fig. 4. Change in Hall–Petch coefficient as a function of the amount of solute carbon or nitrogen in C-series (a) and N-series samples (b).

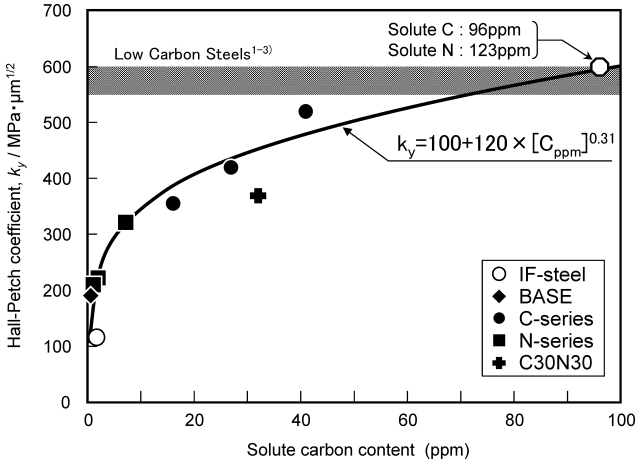


Fig. 5. Change in Hall–Petch coefficient as a function of solute carbon content.

(1) Nitrogen increases the twin density that appears as the increased Hall–Petch coefficient.[8,10]
(2) Planar slip of dislocations promoted by nitrogen enhances the grain refinement strengthening.[8,11–13]
(3) Grain boundary segregation of nitrogen atoms stabilizes the ledge structure of grain boundary working as a dislocation source.[12]
(4) Dislocations are locked by a strong interaction between nitrogen atoms and dislocations.[12]

However, each mechanism was only demonstrated indirectly in a certain kind of steel, and there are also counterarguments for each mechanism. Although we have never reached a consensus, the most frequently used idea, including nonferrous materials, would be the (3) grain boundary segregation theory. This is based on the yielding mechanism proposed by Li[14] that the grain boundary ledges are supposed to be dislocation sources. As representative examples supporting this mechanism, a result in Ni–S alloy by Floreen[15] and that in Fe–P alloy by Spitzig[16] were reported. Both of the results concluded an increase in Hall–Petch coefficient with enhancing the grain boundary segregation.

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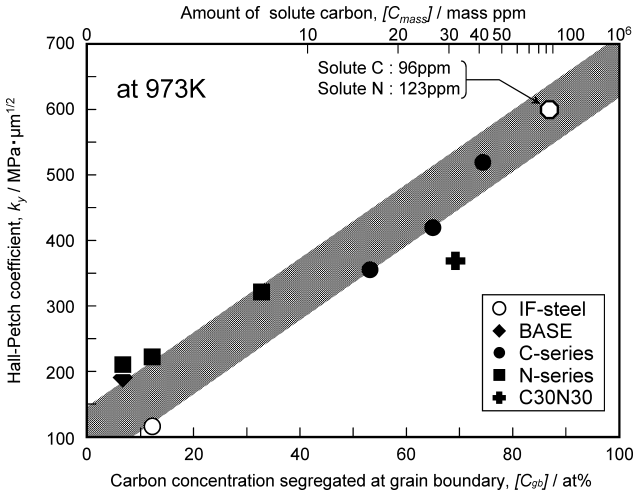


Fig. 6. Reration between Hall–Petch coefficient and grain boundary segregation of carbon.

Here, on the assumption that the classical grain boundary segregation theory would be applied to the case of this study, namely, carbon in ferritic steel, the Hall–Petch coefficient was compared with the concentration of carbon at grain boundary. According to the MacLean's theory,[17] solute carbon concentration segregated at ferrite grain boundary at 973 K is calculated by following equation:

$$[C_{gb}]=[\text{at\%C}]\exp(-Q/RT)/(1+[\text{at\%C}]\exp(-Q/RT)) \quad ...(3)$$

In this equation, [at%C], Q and R are the total carbon concentration in solid solution, the activation energy of carbon for grain boundary segregation and the gas constant, respectively. As for the value of Q, 78 kJ/mol estimated by Grabke[18] was taken.

Figure 6 shows relationship between k_y and segregated carbon content calculated by Eq. (3). k_y value tends to increase linearly with increasing segregated carbon content, which suggests that grain boundary segregation of interstitial alloy elements significantly prevents the yielding of ferritic steel. Only 60 ppm carbon causes marked grain boundary segregation to 80 at%, resulting in the enlargement of the k_y from 100 (in IF-steel) to 550 MPa·μm$^{1/2}$. The solubility of carbon was reported to be approximately 60 ppm even at ambient temperature[19] in Fe–C binary alloy. Therefore, it could be understood that the similar k_y values, around 600 MPa·μm$^{1/2}$, have been reported in lots of previous studies on grain refinement strengthening of ferritic steels except for IF-steels. On the other hand, it is known that nitrogen has less tendency to cause grain boundary segregation in steel, as compared to carbon.[20–22] This agrees well with the experimental results that k_y is hardly influenced by nitrogen. The enlarged k_y by nitrogen mentioned above might be due to the large amount of solute nitrogen in austenite matrix leading to the high concentration of nitrogen at grain boundary even though the segregation coefficient is small.

5. Conclusions

(1) IF-steel has a low Hall–Petch coefficient of approximately 100 MPa·μm$^{1/2}$. The Hall–Petch coefficient of ferritic iron monotonously increases with increasing solute carbon content, and it levels off at 550–600 MPa·μm$^{1/2}$ when the solute carbon content reaches 60 ppm. On the other hand, the solute nitrogen hardly affects the Hall–Petch coefficient.

(2) The increase in Hall–Petch coefficient by solute carbon seems to be derived from the grain boundary segregation of carbon atoms. The small effect of solute nitrogen on Hall–Petch coefficient could be explained by the less tendency of nitrogen atoms to cause grain boundary segregation.

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